Certamen №2

(***Date***: June 07, 2024)

1. (***7 points***) A pointlike nonrelativistic particle of mass m scatters in the time-dependent spherically symmetric harmonic potential

$$U(r,\, t) = \frac{k(t)\, r^2}{2}, \quad k(t) = \frac{\kappa}{t^2}, \quad \kappa > 0. \tag{1}$$

   (a) (**2 *points***) Identify all continuous symmetries of the action. Analyze if there are symmetries of type

$$\boldsymbol{r} \to \alpha\, \boldsymbol{r}, \;\; t \to \beta t, \qquad \alpha, \beta = \text{const}$$

   which do not change the equations of motion.

   (b) (**3 *points***) Construct the integrals of motion which might exist according to Noether's theorem. Demonstrate that these quantities are conserved using the equations of motion.

   (c) (**2 *points***) We have seen in lectures that in static (time-independent) Coulomb potential the particle's trajectory always remains in the plane (so-called scattering plane). Analyze if this property remains valid in the time-dependent potential given in Eq. (1). Analyze if the Kepler's laws remain valid in this potential field.

2. (**8 *points***) A ***relativistic*** particle of mass m moves in the spherically symmetric potential $U(r) = \alpha/r + \beta/r^2$.

   (a) (**2 *points***) Write out the lagrangian of the system, and the corresponding equations of motion

   (b) (**2 *points***) Identify all symmetries of the action and construct all integrals of motion which might exist in this problem.

   (c) (**2 *points***) Try to integrate the equations of motion and find trajectories of bound orbits.

   (d) (**2 *points***) Analyze the behavior of your solution in the limit $c \to \infty$. Assuming that c is large yet finite, find explicitly the first relativistic correction to the trajectory.